

Introduction and Motivation

Apple is one of the most influential and defining companies of our generation. Founded in 1976 by budding industrialists Steve Jobs and Steve Wozniak, Apple spent much of its earlier years with a focus on computer and physical hardware development. In 2007, Apple shifted its focus towards the handheld consumer electronic space with the launch of the iPhone. In the years that followed, a new era of “big technology” appeared - the perception of digital communication and consumption of information was forever changed.

Today, Apple remains at the forefront of the technology industry, expanding its presence across the wearables, tablets, and accessories spaces while continuing to be an industry leader in computer & smartphone development, system AI integration, and ecosystem strategy.

It is this continued innovation that makes Apple’s stock-analysis particularly exciting. Beyond the ground-breaking tech launches and ecosystem updates, is there a hidden, underlying process that fuels Apple stock performance? Additionally, Apple’s exposure to major macroeconomic events and dynamic investor expectations over its financial lifetime makes its stock a compelling time-series candidate. In this paper, we will explore Apple stock analysis from a time-series perspective, studying temporal dependencies, seasonal patterns, and forecasting efforts using monthly-adjusted closing prices.

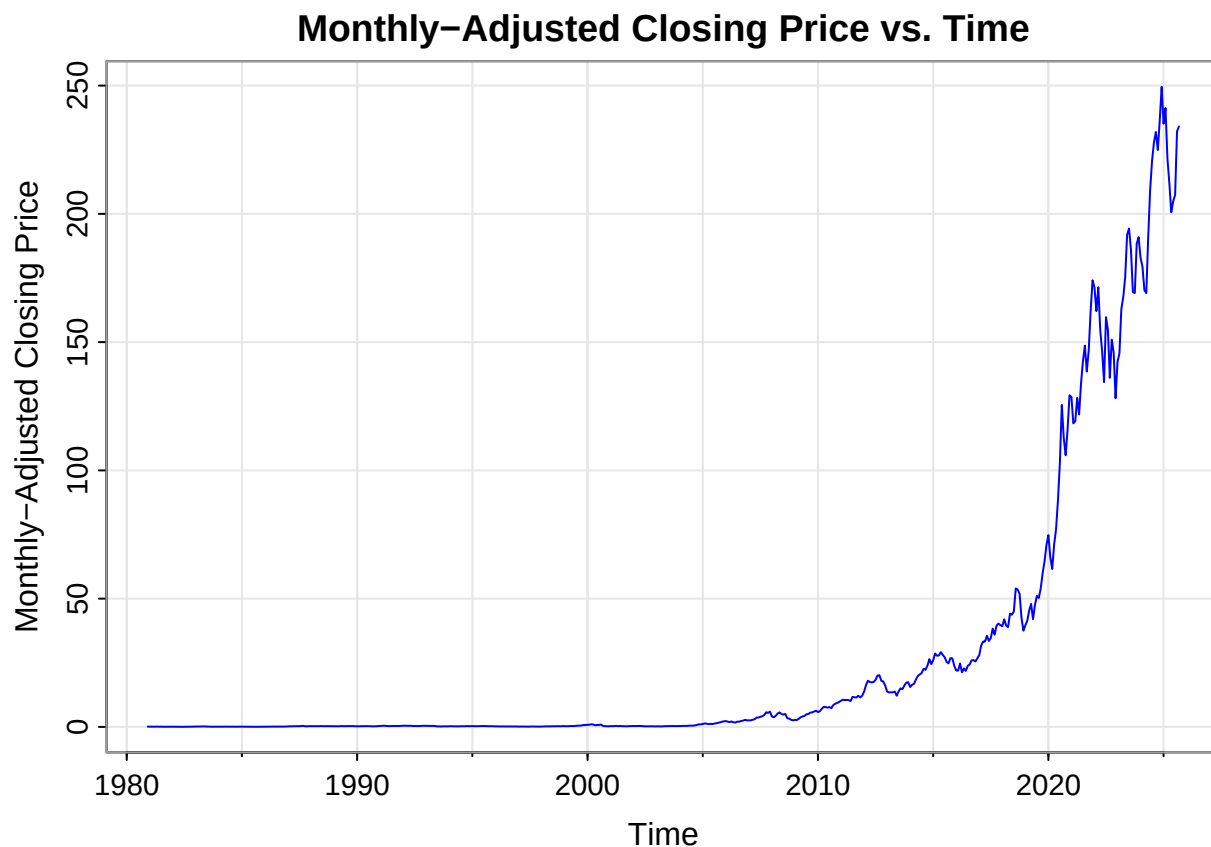
Data Collection

Stock-price data was obtained from a publicly-available Kaggle Dataset. A filtered dataset was obtained by grouping observations by month, and extracting the adjusted close stock price at the end of each month.

A link to the data is provided here: <https://www.kaggle.com/datasets/muhamadatifatif/apple-complete-stocks-datasetall-the-time>

Exploratory Analysis

```
tsplot(apple_data_month, main = "Monthly-Adjusted Closing Price vs. Time",  
       xlab = "Time", ylab = "Monthly-Adjusted Closing Price", col = "blue")
```



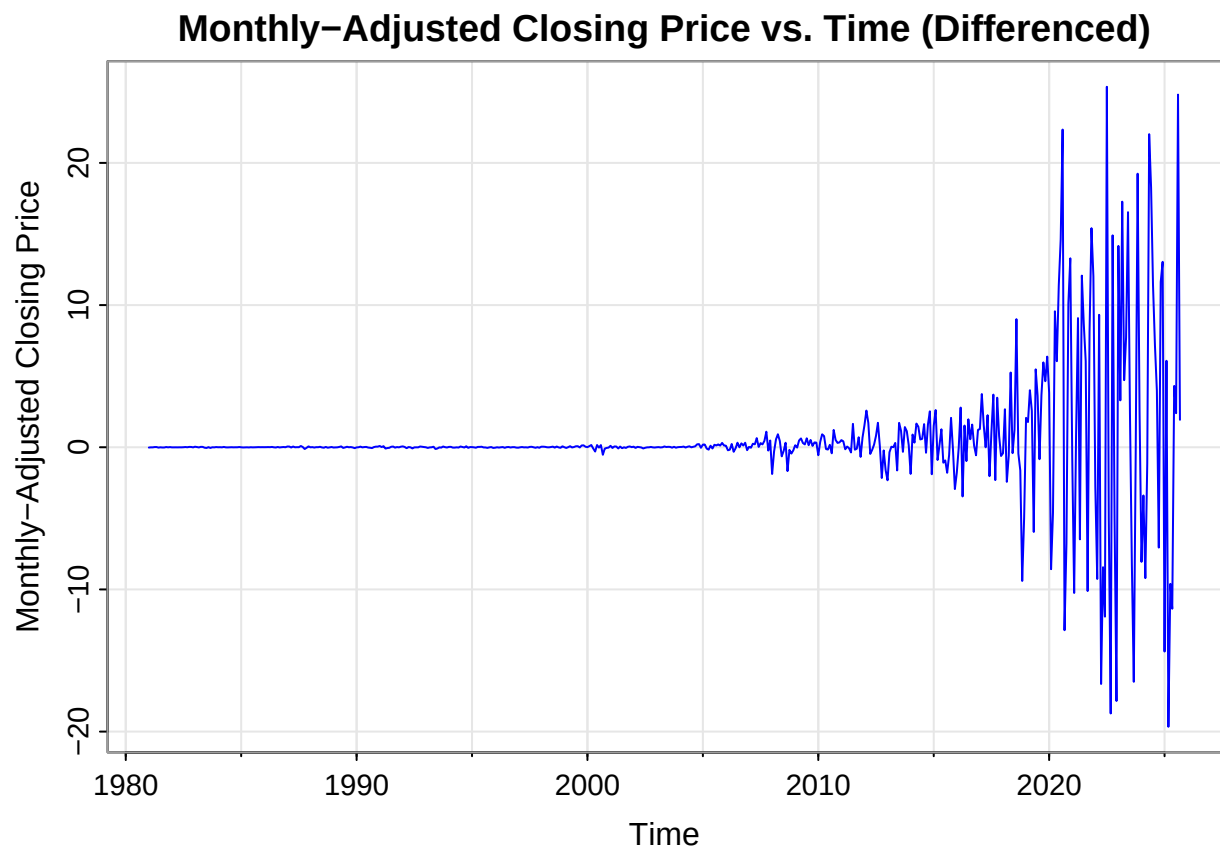
As we model the Monthly-Adjusted Closing Price through time, a clear exponential pattern appears to be exhibited. The rapid rise trend in stock price seems to “begin” roughly around 2010, which corresponds to the time period immediately after the first iPhone launch.

In order to model a time-series effectively, two important assumptions need to be satisfied:

Stationarity: The time-series has a constant mean. Homoskedasticity: The variance of the time-series remains constant through time.

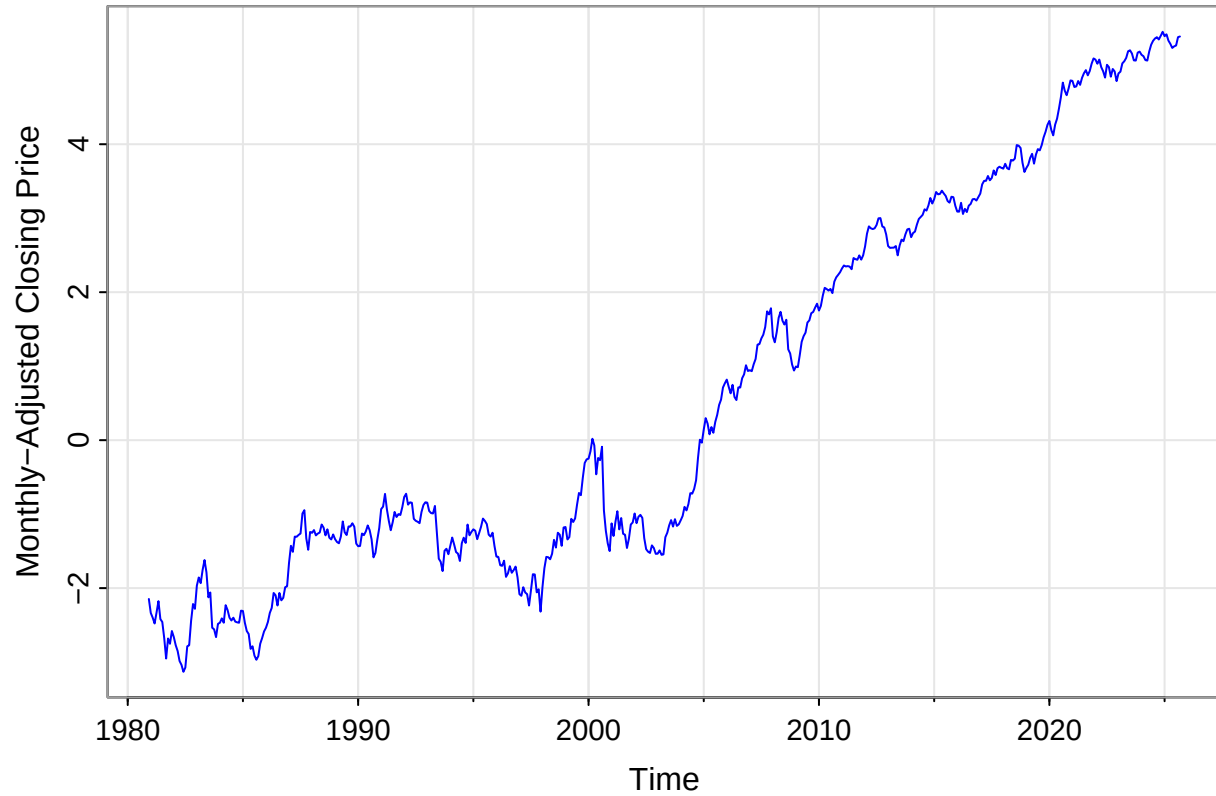
In the plot above, both of these assumptions are violated. Hence, transformations need to be undertaken in order to turn the original time series into an “analysis-ready” version.

```
tsplot(diff(apple_data_month), main =  
      "Monthly-Adjusted Closing Price vs. Time (Differenced)", xlab = "Time",  
      ylab = "Monthly-Adjusted Closing Price", col = "blue")
```

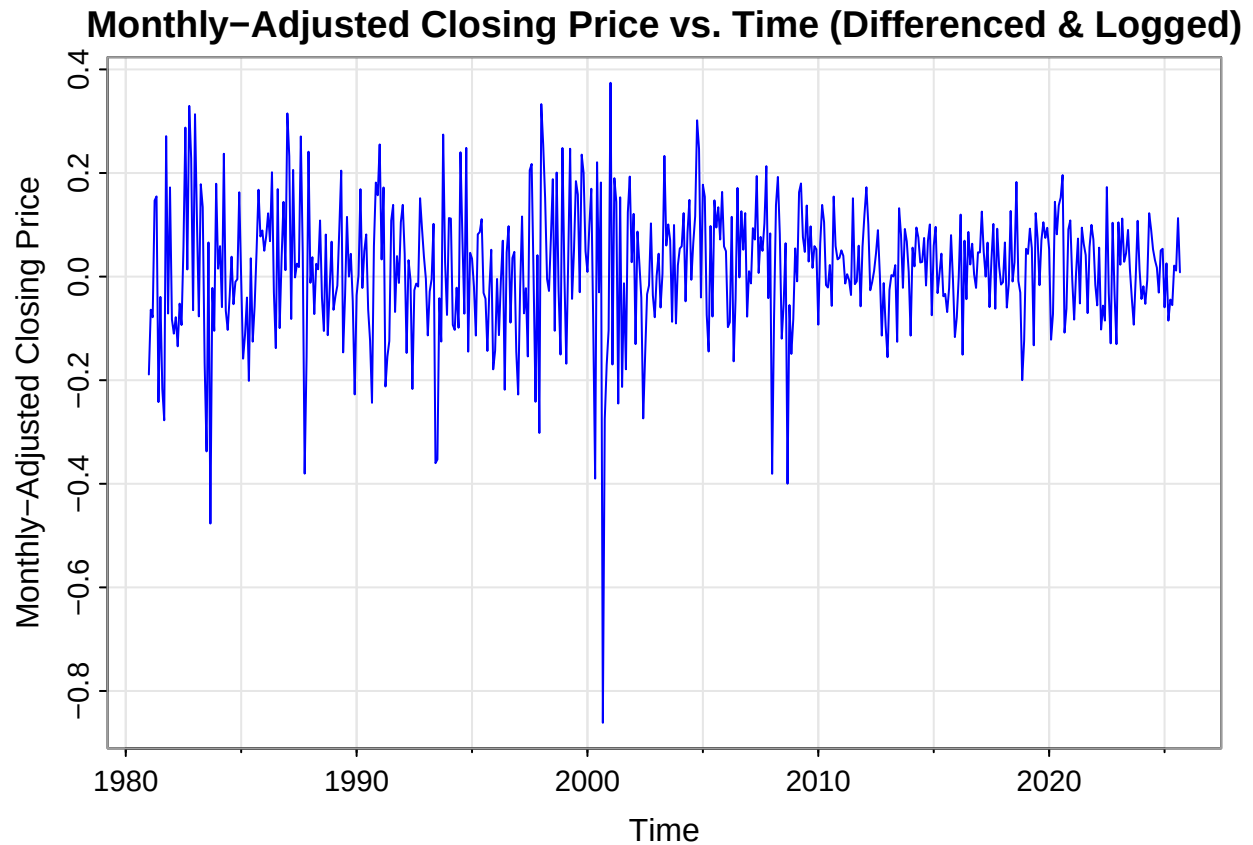


```
tsplot(log(apple_data_month), main =  
    "Monthly-Adjusted Closing Price vs. Time (Logged)", xlab = "Time",  
    ylab = "Monthly-Adjusted Closing Price", col = "blue")
```

Monthly-Adjusted Closing Price vs. Time (Logged)



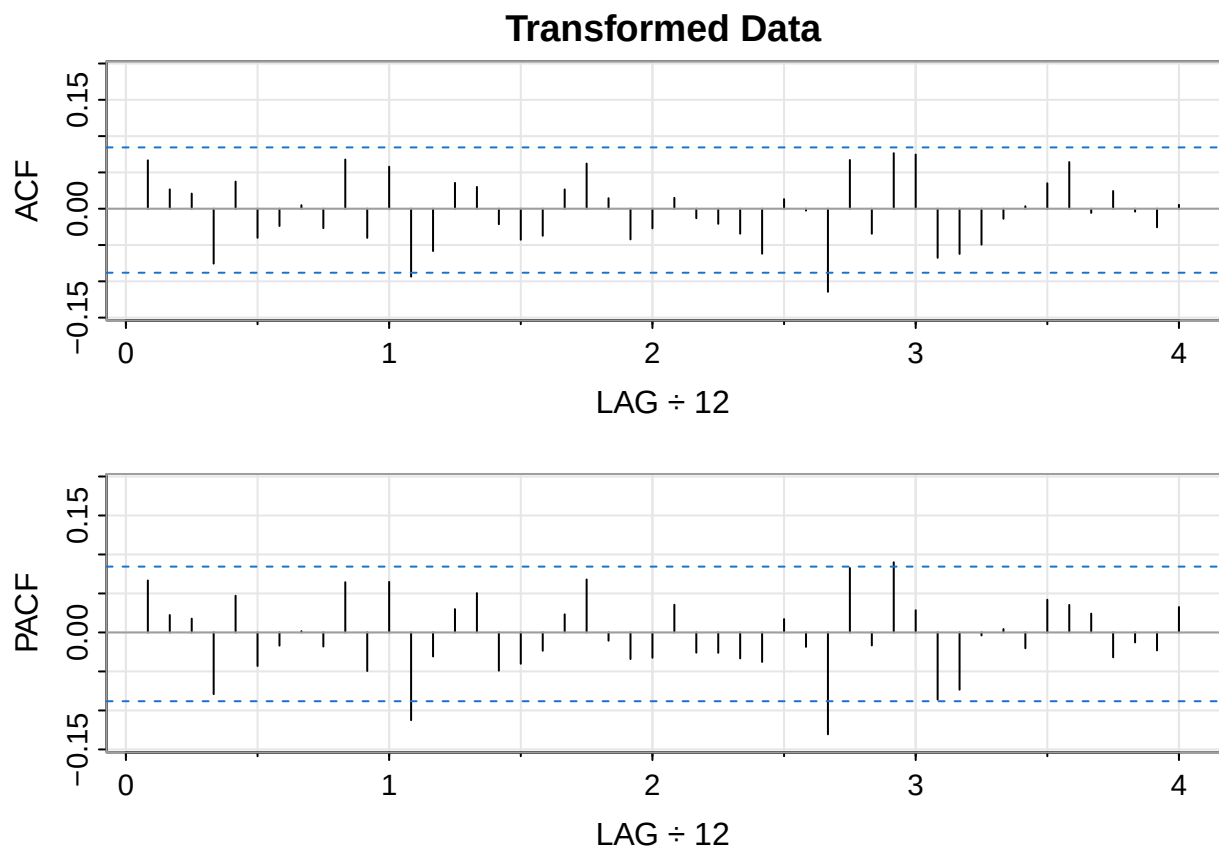
```
tsplot(diff(log(apple_data_month)), main =  
    "Monthly-Adjusted Closing Price vs. Time (Differenced & Logged)",  
    xlab = "Time", ylab = "Monthly-Adjusted Closing Price", col = "blue")
```



In the plots above, 3 different transforming techniques are explored. Differencing and logging was undertaken, but ultimately a combination approach yielded a series better satisfying the homoskedasticity and stationarity assumptions.

ACF and PACF Interpretations

```
apple_data_transformed <- diff(log(apple_data_month))  
  
acf2(apple_data_transformed, main = "Transformed Data")
```



```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
## ACF  0.07 0.03 0.02 -0.08 0.04 -0.04 -0.02  0 -0.03  0.07 -0.04  0.06 -0.09
## PACF 0.07 0.02 0.02 -0.08 0.05 -0.04 -0.02  0 -0.02  0.06 -0.05  0.06 -0.11
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF -0.06  0.04  0.03 -0.02 -0.04 -0.04  0.03  0.06  0.01 -0.04 -0.03  0.01
## PACF -0.03  0.03  0.05 -0.05 -0.04 -0.02  0.02  0.07 -0.01 -0.03 -0.03  0.04
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF -0.01 -0.02 -0.03 -0.06  0.01  0.00 -0.11  0.07 -0.03  0.08  0.07 -0.07
## PACF -0.03 -0.03 -0.03 -0.04  0.02 -0.02 -0.13  0.08 -0.02  0.09  0.03 -0.09
##      [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48]
## ACF -0.06 -0.05 -0.01  0.00  0.04  0.06 -0.01  0.02  0.00 -0.03  0.01
## PACF -0.07  0.00  0.00 -0.02  0.04  0.04  0.02 -0.03 -0.01 -0.02  0.03
```

The ACF and PACF plots are designed to show autocorrelation between data points and previous data points at a certain lag distance h . The dotted-blue lines are visual indicators showing significance thresholds.

From the plots above, the ACF plot shows ~2 lags that appear to be significant, while the PACF plot shows ~3 lags that appear to be significant. From a cumulative glance, the ACF or PACF plots do not seem to exhibit any tailing or cutting off.

However, there may be visual representation of a more nuanced underlying process. Significant lags seem to appear at roughly regular time intervals. When treating these rough time intervals as “pseudo time-series”, there are some tailing/cutting off patterns exhibited here, suggesting that this series has a seasonality component.

Model Fitting & Diagnostics

First, we will consider the efficacy of simpler models. To begin, AR and MA models will be fit.

AR and MA models can be described as such:

AR (Autoregressive Models): Values at a future time t can be predicted using a linear combination of previous time values.

MA (Moving Average Models): Values at a future time t can be predicted using previous forecast errors (or shocks).

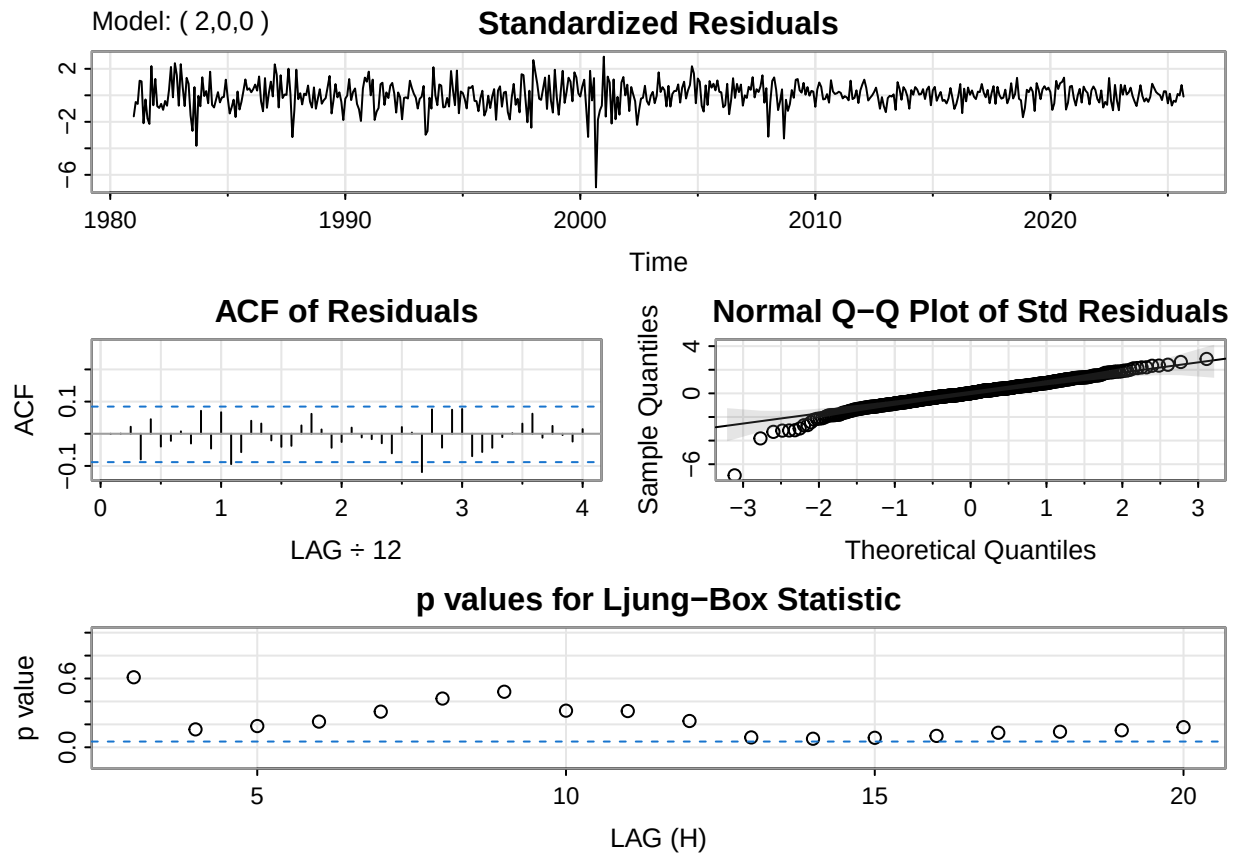
AR Models

```
par(mfrow = c(3, 1))
ar1_model <- sarima(apple_data_transformed, 1, 0, 0)

## initial value -2.058217
## iter 2 value -2.060457
## iter 3 value -2.060457
## iter 4 value -2.060459
## iter 5 value -2.060460
## iter 5 value -2.060460
## iter 5 value -2.060460
## final value -2.060460
## converged
## initial value -2.059023
## iter 2 value -2.059025
## iter 3 value -2.059028
## iter 3 value -2.059028
## iter 3 value -2.059028
## final value -2.059028
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0669 0.0431   1.551  0.1215
## xmean     0.0141 0.0059   2.395  0.0170
##
## sigma^2 estimated as 0.01627598 on 535 degrees of freedom
##
## AIC = -1.269006  AICc = -1.268964  BIC = -1.245062
##
```



```
## ar1      0.0654 0.0432 1.5139 0.1306
## ar2      0.0225 0.0432 0.5206 0.6029
## xmean    0.0141 0.0060 2.3403 0.0196
##
## sigma^2 estimated as 0.01626774 on 534 degrees of freedom
##
## AIC = -1.265786 AICc = -1.265703 BIC = -1.233861
##
```



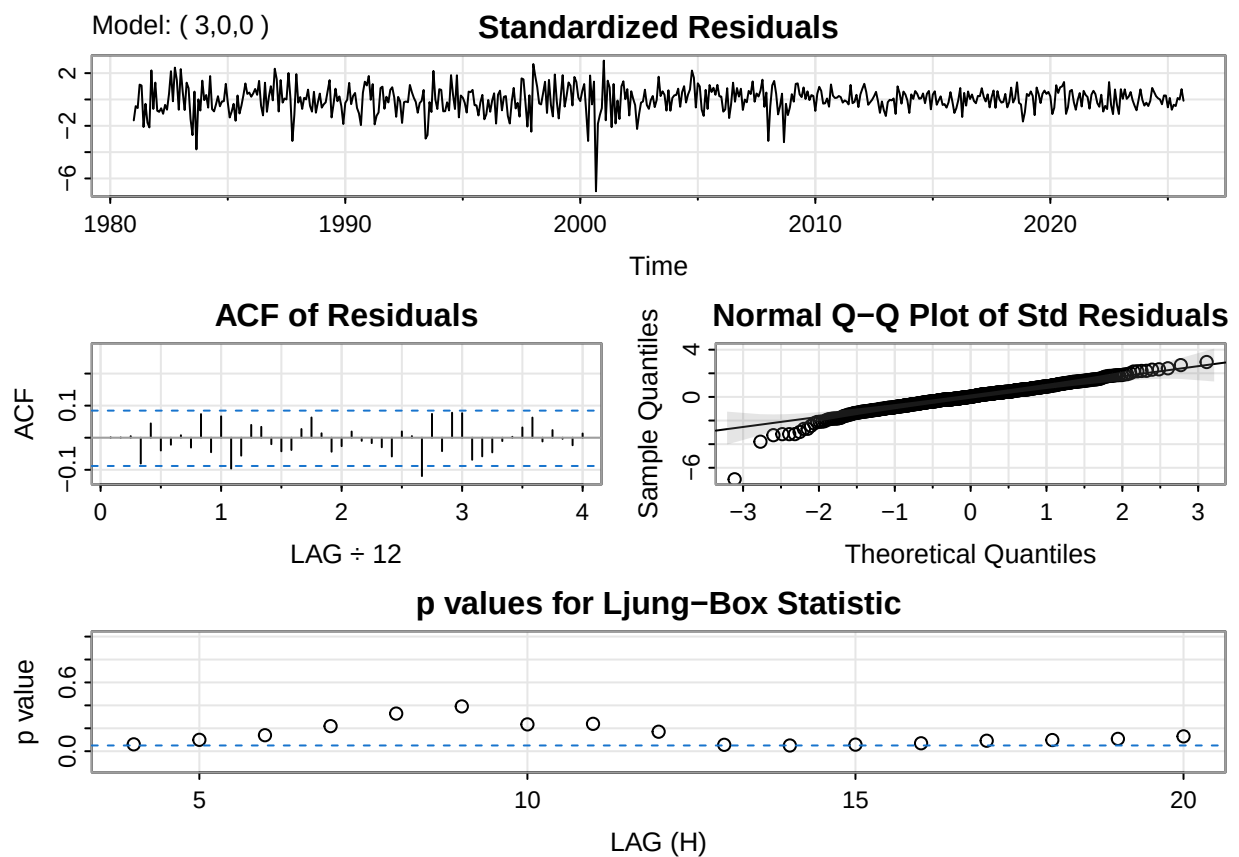
```
ar3_model <- sarima(apple_data_transformed, 3, 0, 0)
```

```
## initial value -2.057188
## iter 2 value -2.059622
## iter 3 value -2.059647
## iter 4 value -2.059649
## iter 5 value -2.059654
## iter 6 value -2.059654
## iter 6 value -2.059654
## iter 6 value -2.059654
## final value -2.059654
## converged
## initial value -2.059420
## iter 2 value -2.059429
## iter 3 value -2.059434
## iter 4 value -2.059439
## iter 5 value -2.059440
## iter 6 value -2.059440
```

```

## iter    6 value -2.059440
## iter    6 value -2.059440
## final   value -2.059440
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0650 0.0432  1.5053  0.1328
## ar2      0.0214 0.0433  0.4934  0.6219
## ar3      0.0179 0.0432  0.4144  0.6787
## xmean     0.0141 0.0061  2.2971  0.0220
##
## sigma^2 estimated as 0.01626251 on 533 degrees of freedom
##
## AIC = -1.262382  AICc = -1.262242  BIC = -1.222475
##

```



AR Model Diagnostics

AR1 Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). Most importantly, the p-values described in the Ljung-Box Statistic are all beyond the blue significance threshold depicted, indicating that sufficient de-trending was undertaken. In other words, the model captures the complexities of the data well and does not leave any further complexities unexplained.

AR2 Model: The standardized residuals appear to roughly follow the rules of stationarity and constant

variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). The p-values described in the Ljung-Box Statistic are largely beyond the blue significance threshold depicted, with a couple points right on the threshold. For the most part, this indicates that sufficient de-trending was undertaken. The model captures the complexities of the data largely well and does not leave any further complexities unexplained.

AR3 Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). However, several p-values described in the Ljung-Box Statistic lie on or below the blue threshold depicted, indicating that insufficient de-trending was undertaken. In other words, the model captures some of the complexities of the data, but leaves further complexities unexplained.

MA Models

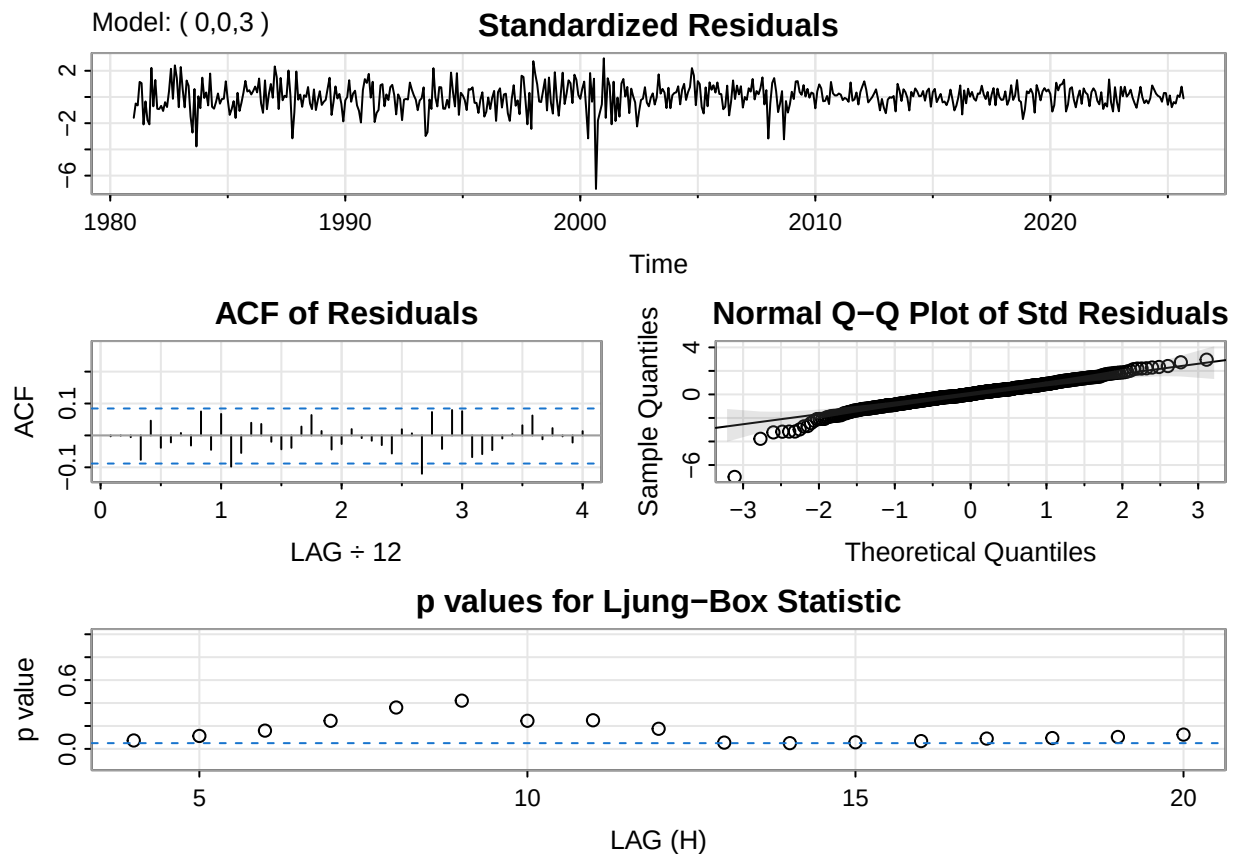
```
par(mfrow = c(3, 1))
ma1_model <- sarima(apple_data_transformed, 0, 0, 1)

## initial value -2.056794
## iter 2 value -2.058920
## iter 3 value -2.058924
## iter 3 value -2.058924
## iter 3 value -2.058924
## final value -2.058924
## converged
## initial value -2.058930
## iter 1 value -2.058930
## final value -2.058930
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          0.0640 0.0422  1.5160  0.1301
## xmean        0.0141 0.0059  2.4142  0.0161
##
## sigma^2 estimated as 0.0162792 on 535 degrees of freedom
##
## AIC = -1.268809 AICc = -1.268767 BIC = -1.244865
##
```


The figure displays four diagnostic plots for the Ljung-Box model:

- Standardized Residuals:** A time series plot showing residuals from 1980 to 2020. The residuals fluctuate around zero, with a notable sharp negative spike around 2000.
- ACF of Residuals:** A plot of the Autocorrelation Function (ACF) for lags 0 to 40. The autocorrelation is mostly within the 95% confidence interval (dashed blue lines), indicating no significant linear structure.
- Normal Q-Q Plot of Std Residuals:** A plot comparing sample quantiles against theoretical quantiles. The points closely follow the diagonal line, suggesting the residuals are approximately normally distributed.
- p values for Ljung-Box Statistic:** A plot showing the p-values for the Ljung-Box test across lags 1 to 20. Most p-values are above the 0.05 significance level (dashed blue line), indicating no significant autocorrelation.

```
##
## AIC = -1.262941  AICc = -1.262801  BIC = -1.223034
##
```



MA Model Diagnostics:

MA1 Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). Most importantly, the p-values described in the Ljung-Box Statistic are all beyond the blue significance threshold depicted, indicating that sufficient de-trending was undertaken. In other words, the model captures the complexities of the data well and does not leave any further complexities unexplained.

MA2 Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). The p-values described in the Ljung-Box Statistic are largely beyond the blue significance threshold depicted, with a couple points right on the threshold. For the most part, this indicates that sufficient de-trending was undertaken. The model captures the complexities of the data largely well and does not leave any further complexities unexplained.

MA3 Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). However, several p-values described in the Ljung-Box Statistic lie on or below the blue threshold depicted, indicating that insufficient de-trending was undertaken. In other words, the model captures some of the complexities of the data, but leaves further complexities unexplained.

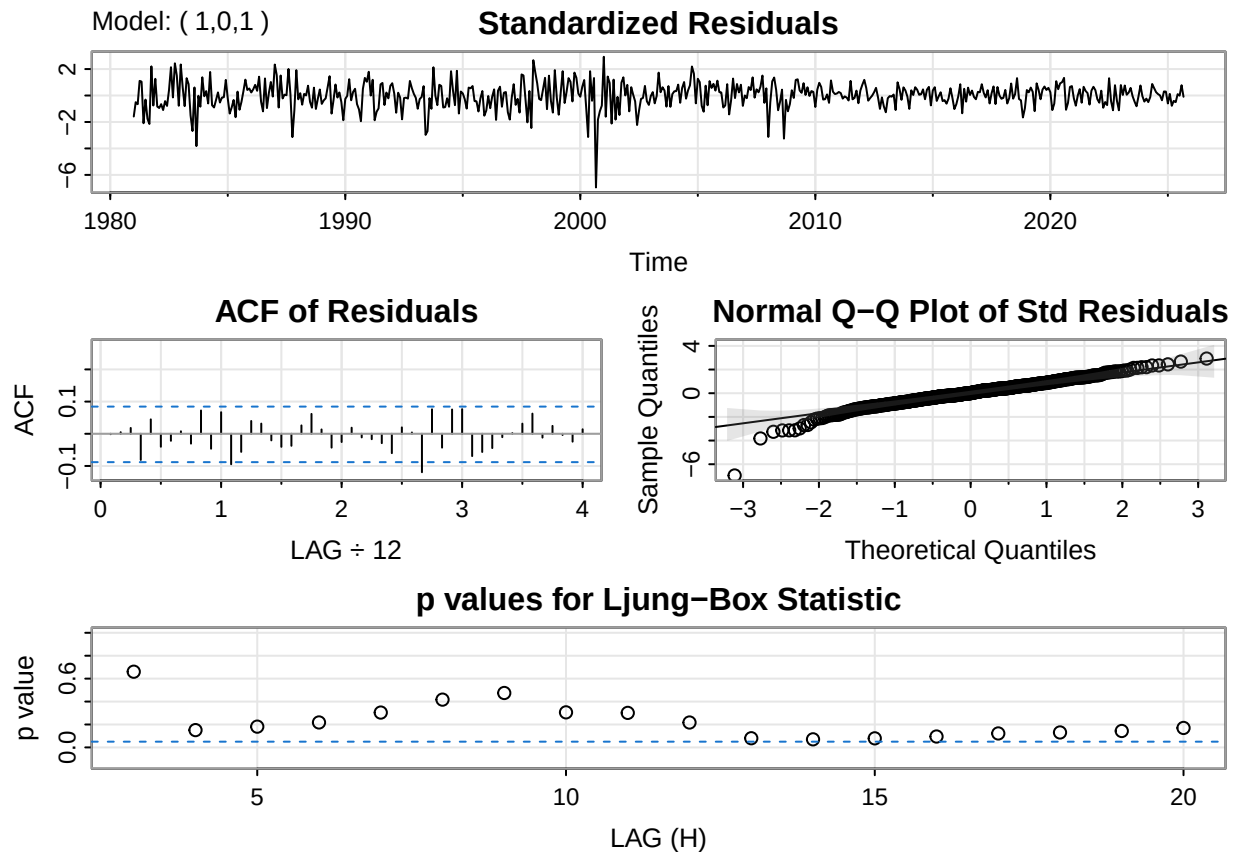
Ironically, we see that the more complex models in both the AR and MA case actually fit more *poorly* than the simpler models. From above, we clearly see that the AR1 and MA1 models outshine the rest in terms of

We will now proceed to fitting ARMA models. An ARMA model can be described as such:

ARMA Models

```
## initial value -2.058217
## iter 2 value -2.059608
## iter 3 value -2.060364
## iter 4 value -2.060369
## iter 5 value -2.060507
## iter 6 value -2.060541
## iter 7 value -2.060755
## iter 8 value -2.060881
## iter 9 value -2.060906
## iter 10 value -2.060968
## iter 11 value -2.060969
## iter 12 value -2.060975
## iter 13 value -2.060980
## iter 14 value -2.060981
## iter 15 value -2.060981
## iter 15 value -2.060981
## final value -2.060981
## converged
## initial value -2.059241
## iter 2 value -2.059244
## iter 3 value -2.059247
## iter 4 value -2.059252
## iter 5 value -2.059252
## iter 6 value -2.059253
## iter 7 value -2.059253
## iter 8 value -2.059253
## iter 9 value -2.059253
## iter 10 value -2.059253
## iter 11 value -2.059253
## iter 12 value -2.059253
## iter 13 value -2.059254
## iter 14 value -2.059254
## iter 15 value -2.059254
## iter 16 value -2.059254
## iter 17 value -2.059255
## iter 18 value -2.059255
## iter 18 value -2.059255
## iter 18 value -2.059255
## final value -2.059255
## converged
## <><><><><><><><><><>
##
## Coefficients:
## Estimate SE t.value p.value
## ar1 0.3269 0.4057 0.8058 0.4207
```

```
## ma1      -0.2603 0.4144 -0.6281 0.5302
## xmean     0.0141 0.0060 2.3331 0.0200
##
## sigma^2 estimated as 0.01626857 on 534 degrees of freedom
##
## AIC = -1.265736 AICc = -1.265652 BIC = -1.23381
##
```



```
arma2_model <- sarima(apple_data_transformed, 2, 0, 2)
```

```
## initial value -2.057632
## iter 2 value -2.059177
## iter 3 value -2.059975
## iter 4 value -2.059978
## iter 5 value -2.059981
## iter 6 value -2.059989
## iter 7 value -2.059994
## iter 8 value -2.059999
## iter 9 value -2.060004
## iter 10 value -2.060015
## iter 11 value -2.060032
## iter 12 value -2.060049
## iter 13 value -2.060052
## iter 14 value -2.060055
## iter 15 value -2.060056
## iter 16 value -2.060058
## iter 17 value -2.060058
```



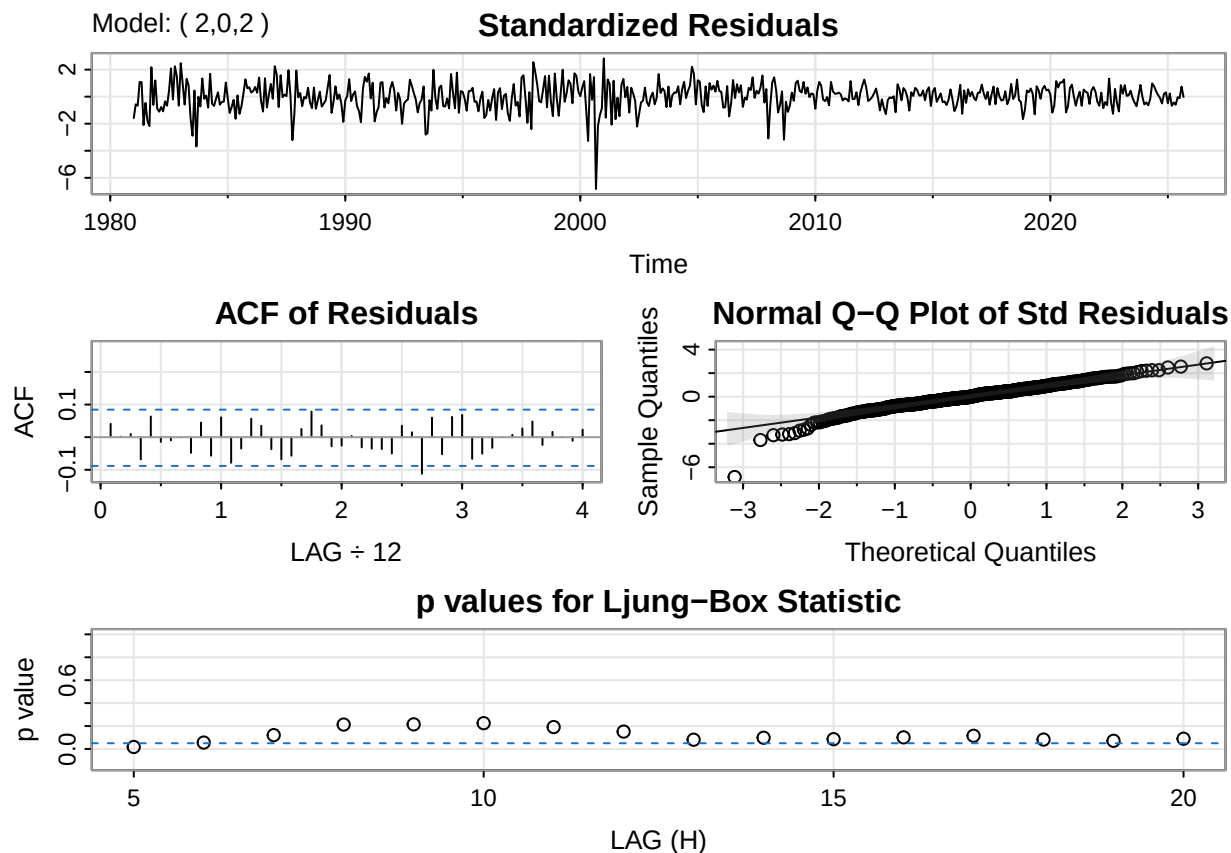
```
## iter 18 value -2.060059
## iter 19 value -2.060059
## iter 20 value -2.060061
## iter 21 value -2.060067
## iter 22 value -2.060108
## iter 23 value -2.060113
## iter 24 value -2.060188
## iter 25 value -2.060191
## iter 26 value -2.060194
## iter 27 value -2.060198
## iter 28 value -2.060205
## iter 29 value -2.060212
## iter 30 value -2.060223
## iter 31 value -2.060241
## iter 32 value -2.060288
## iter 33 value -2.060374
## iter 34 value -2.060419
## iter 35 value -2.060499
## iter 36 value -2.060564
## iter 37 value -2.060604
## iter 38 value -2.060613
## iter 39 value -2.060615
## iter 40 value -2.060639
## iter 41 value -2.060681
## iter 42 value -2.060791
## iter 43 value -2.060946
## iter 44 value -2.061053
## iter 45 value -2.061079
## iter 46 value -2.061092
## iter 47 value -2.061146
## iter 48 value -2.061190
## iter 49 value -2.061300
## iter 50 value -2.061378
## iter 51 value -2.061451
## iter 52 value -2.061512
## iter 53 value -2.061577
## iter 54 value -2.061628
## iter 55 value -2.061653
## iter 56 value -2.061656
## iter 57 value -2.061679
## iter 58 value -2.061749
## iter 59 value -2.061839
## iter 60 value -2.061903
## iter 61 value -2.061991
## iter 62 value -2.062093
## iter 63 value -2.062255
## iter 64 value -2.062431
## iter 65 value -2.062753
## iter 66 value -2.063329
## iter 67 value -2.063412
## iter 68 value -2.064057
## iter 69 value -2.065076
## iter 70 value -2.065114
## iter 71 value -2.065137
```

```
## iter 72 value -2.065214
## iter 73 value -2.065239
## iter 74 value -2.065319
## iter 75 value -2.065372
## iter 76 value -2.065410
## iter 77 value -2.065489
## iter 78 value -2.065529
## iter 79 value -2.065543
## iter 80 value -2.065595
## iter 81 value -2.065662
## iter 82 value -2.065694
## iter 83 value -2.065725
## iter 84 value -2.065773
## iter 85 value -2.065822
## iter 86 value -2.065881
## iter 87 value -2.065893
## iter 88 value -2.065893
## iter 88 value -2.065893
## final value -2.065893
## converged
## initial value -2.058304
## iter 2 value -2.058641
## iter 3 value -2.059110
## iter 4 value -2.059408
## iter 5 value -2.059644
## iter 6 value -2.059842
## iter 7 value -2.059982
## iter 8 value -2.060024
## iter 9 value -2.060024
## iter 10 value -2.060025
## iter 11 value -2.060026
## iter 12 value -2.060027
## iter 13 value -2.060027
## iter 14 value -2.060027
## iter 15 value -2.060028
## iter 16 value -2.060029
## iter 17 value -2.060029
## iter 18 value -2.060031
## iter 19 value -2.060033
## iter 20 value -2.060053
## iter 21 value -2.060056
## iter 22 value -2.060061
## iter 23 value -2.060073
## iter 24 value -2.060082
## iter 25 value -2.060084
## iter 26 value -2.060086
## iter 27 value -2.060090
## iter 28 value -2.060102
## iter 29 value -2.060124
## iter 30 value -2.060162
## iter 31 value -2.060252
## iter 32 value -2.060274
## iter 33 value -2.060302
## iter 34 value -2.060367
```

```

## iter 35 value -2.060389
## iter 36 value -2.060408
## iter 37 value -2.060421
## iter 38 value -2.060435
## iter 39 value -2.060449
## iter 40 value -2.060478
## iter 41 value -2.060546
## iter 42 value -2.060638
## iter 43 value -2.060664
## iter 44 value -2.060732
## iter 45 value -2.060861
## iter 46 value -2.060895
## iter 47 value -2.060912
## iter 48 value -2.060935
## iter 49 value -2.060940
## iter 50 value -2.060950
## iter 51 value -2.060969
## iter 52 value -2.061021
## iter 53 value -2.061100
## iter 54 value -2.061118
## iter 55 value -2.061124
## iter 56 value -2.061141
## iter 57 value -2.061158
## iter 58 value -2.061163
## iter 59 value -2.061166
## iter 60 value -2.061166
## iter 61 value -2.061167
## iter 62 value -2.061167
## iter 63 value -2.061167
## iter 64 value -2.061168
## iter 64 value -2.061168
## final value -2.061168
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      1.4461 0.0464 31.1915 0.0000
## ar2     -0.9728 0.0287 -33.8800 0.0000
## ma1     -1.4261 0.0350 -40.7194 0.0000
## ma2      0.9674 0.0451 21.4726 0.0000
## xmean    0.0140 0.0056  2.4846 0.0133
##
## sigma^2 estimated as 0.01619754 on 532 degrees of freedom
##
## AIC = -1.262113  AICc = -1.261902  BIC = -1.214224
##

```



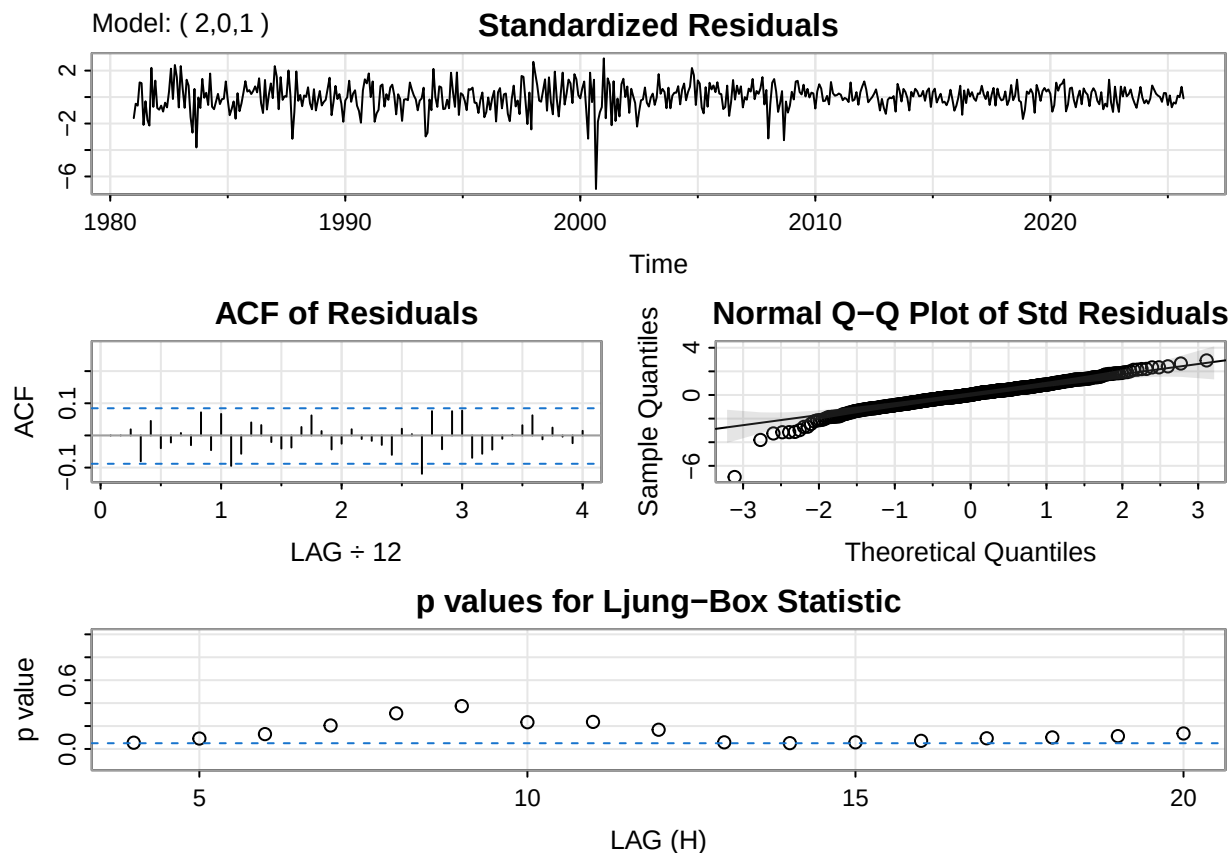
```
arma21_model <- sarima(apple_data_transformed, 2, 0, 1)
```

```
## initial value -2.057632
## iter 2 value -2.057785
## iter 3 value -2.059980
## iter 4 value -2.059986
## iter 5 value -2.059988
## iter 6 value -2.059995
## iter 7 value -2.059997
## iter 8 value -2.060000
## iter 9 value -2.060006
## iter 10 value -2.060021
## iter 11 value -2.060030
## iter 12 value -2.060034
## iter 13 value -2.060038
## iter 14 value -2.060042
## iter 15 value -2.060043
## iter 16 value -2.060049
## iter 17 value -2.060059
## iter 18 value -2.060076
## iter 19 value -2.060093
## iter 20 value -2.060098
## iter 21 value -2.060100
## iter 22 value -2.060101
## iter 23 value -2.060102
## iter 24 value -2.060102
```

```

## iter 24 value -2.060102
## iter 24 value -2.060102
## final value -2.060102
## converged
## initial value -2.059288
## iter 2 value -2.059291
## iter 3 value -2.059295
## iter 4 value -2.059299
## iter 5 value -2.059300
## iter 6 value -2.059300
## iter 6 value -2.059300
## iter 6 value -2.059300
## final value -2.059300
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1968 0.7240  0.2718  0.7859
## ar2      0.0144 0.0687  0.2090  0.8345
## ma1     -0.1314 0.7229 -0.1818  0.8558
## xmean     0.0141 0.0061  2.3296  0.0202
##
## sigma^2 estimated as 0.01626709 on 533 degrees of freedom
##
## AIC = -1.262102  AICc = -1.261962  BIC = -1.222195
##

```



```
arma12_model <- sarima(apple_data_transformed, 1, 0, 2)
```

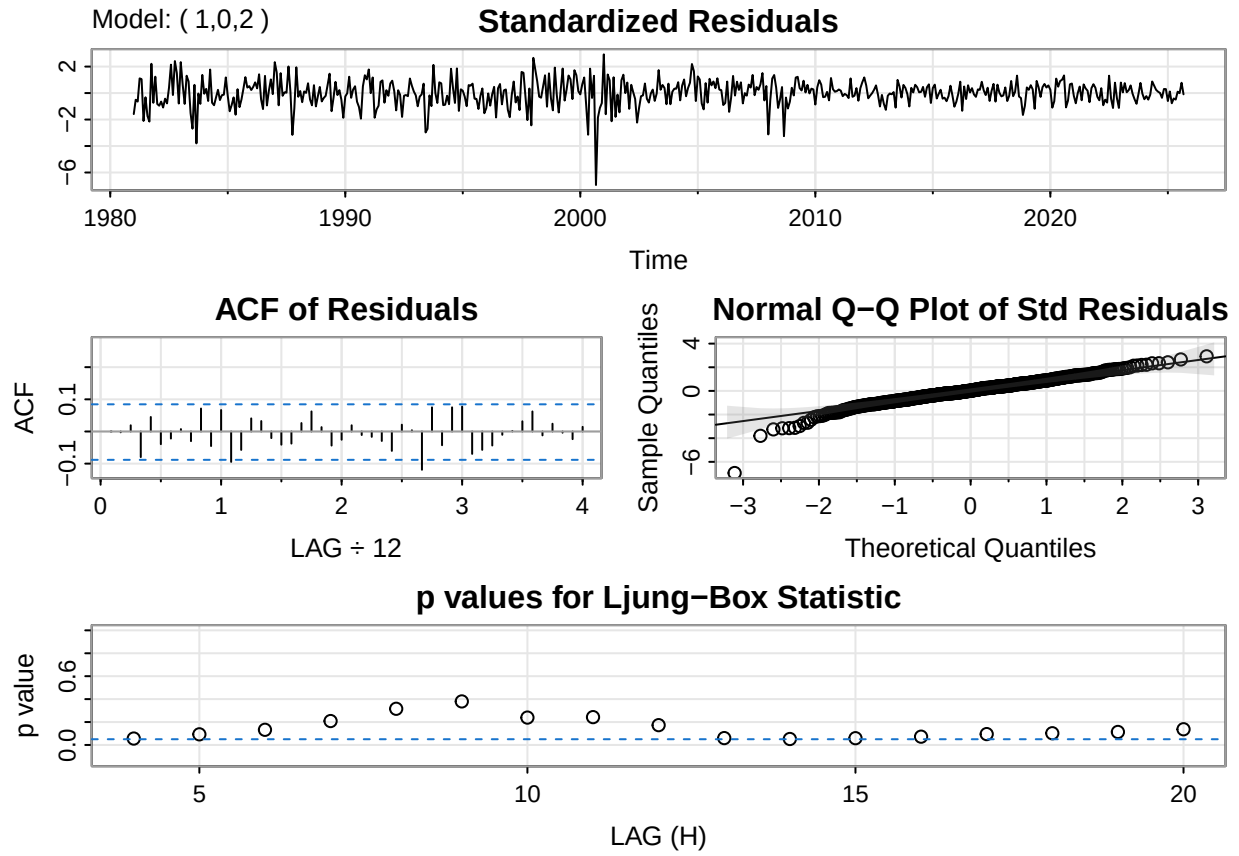
```
## initial value -2.058217
## iter 2 value -2.058253
## iter 3 value -2.060602
## iter 4 value -2.060606
## iter 5 value -2.060621
## iter 6 value -2.060644
## iter 7 value -2.060679
## iter 8 value -2.060720
## iter 9 value -2.060770
## iter 10 value -2.060834
## iter 11 value -2.060859
## iter 12 value -2.060897
## iter 13 value -2.060941
## iter 14 value -2.060947
## iter 15 value -2.060948
## iter 16 value -2.060949
## iter 17 value -2.060953
## iter 18 value -2.060961
## iter 19 value -2.060975
## iter 20 value -2.060981
## iter 21 value -2.060985
## iter 22 value -2.060986
## iter 23 value -2.060988
## iter 24 value -2.060988
```

```
## iter 25 value -2.060988
## iter 26 value -2.060989
## iter 27 value -2.060990
## iter 28 value -2.060992
## iter 29 value -2.060993
## iter 30 value -2.060994
## iter 31 value -2.060994
## iter 32 value -2.060994
## iter 33 value -2.060994
## iter 34 value -2.060995
## iter 35 value -2.060995
## iter 36 value -2.060996
## iter 37 value -2.060996
## iter 38 value -2.060997
## iter 39 value -2.060997
## iter 40 value -2.060997
## iter 41 value -2.060997
## iter 41 value -2.060997
## iter 41 value -2.060997
## final value -2.060997
## converged
## initial value -2.059272
## iter 2 value -2.059275
## iter 3 value -2.059278
## iter 4 value -2.059283
## iter 5 value -2.059284
## iter 6 value -2.059284
## iter 7 value -2.059285
## iter 8 value -2.059286
## iter 9 value -2.059288
## iter 10 value -2.059294
## iter 11 value -2.059295
## iter 12 value -2.059299
## iter 13 value -2.059299
## iter 14 value -2.059301
## iter 15 value -2.059301
## iter 16 value -2.059301
## iter 17 value -2.059302
## iter 18 value -2.059303
## iter 19 value -2.059306
## iter 20 value -2.059307
## iter 21 value -2.059309
## iter 22 value -2.059309
## iter 23 value -2.059310
## iter 24 value -2.059310
## iter 25 value -2.059310
## iter 26 value -2.059310
## iter 27 value -2.059311
## iter 28 value -2.059313
## iter 29 value -2.059313
## iter 30 value -2.059314
## iter 31 value -2.059314
## iter 32 value -2.059315
## iter 33 value -2.059315
```

```

## iter 34 value -2.059315
## iter 35 value -2.059315
## iter 36 value -2.059316
## iter 37 value -2.059317
## iter 38 value -2.059317
## iter 39 value -2.059318
## iter 40 value -2.059318
## iter 41 value -2.059318
## iter 42 value -2.059318
## iter 43 value -2.059319
## iter 44 value -2.059319
## iter 45 value -2.059319
## iter 46 value -2.059320
## iter 47 value -2.059320
## iter 48 value -2.059320
## iter 49 value -2.059320
## iter 50 value -2.059320
## iter 51 value -2.059320
## iter 52 value -2.059320
## iter 53 value -2.059320
## iter 54 value -2.059321
## iter 55 value -2.059321
## iter 56 value -2.059321
## iter 57 value -2.059322
## iter 58 value -2.059322
## iter 59 value -2.059322
## iter 59 value -2.059322
## iter 59 value -2.059322
## final value -2.059322
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1971 0.6385  0.3088  0.7576
## ma1     -0.1322 0.6372 -0.2075  0.8357
## ma2      0.0171 0.0675  0.2539  0.7997
## xmean    0.0141 0.0061  2.3254  0.0204
##
## sigma^2 estimated as 0.01626639 on 533 degrees of freedom
##
## AIC = -1.262144  AICc = -1.262004  BIC = -1.222238
##

```

ARMA Model Diagnostics:

ARMA (1, 1) Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). The p-values described in the Ljung-Box Statistic are all beyond the blue significance threshold depicted, except for a couple. This indicates that sufficient de-trending was undertaken. In other words, the model captures the complexities of the data largely well and does not leave any further complexities unexplained.

ARMA (2, 2) Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). However, many p-values described in the Ljung-Box Statistic lie on or below the blue threshold depicted, indicating that insufficient de-trending was undertaken. In other words, the model captures some of the complexities of the data, but leaves further complexities unexplained.

ARMA (2, 1) Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). However, several p-values described in the Ljung-Box Statistic lie on or below the blue threshold depicted, indicating that insufficient de-trending was undertaken. In other words, the model captures some of the complexities of the data, but leaves further complexities unexplained.

ARMA (1, 2) Model: The standardized residuals appear to roughly follow the rules of stationarity and constant variance. The ACF plot does not show any consistently significant lags, and the Normal-QQ plot show the errors following the line of normality well (barring some deviation in the tails). However, several p-values described in the Ljung-Box Statistic lie on or below the blue threshold depicted, indicating that insufficient de-trending was undertaken. In other words, the model captures some of the complexities of the

data, but leaves further complexities unexplained.

From the following, the ARMA(1, 1) Model fits the data best.

Model Selection

From the previous section, we have found the the AR1, AR2, MA1, MA2, and ARMA(1, 1) models to fit the data best. In regards to comparing model fit against one another, we turn to another fit assessment: AIC (Akaike Information Criteria), AICc (Corrected Akaike Information Criteria), and BIC (Bayesian Information Criteria).

These selection criteria attempt to define the tradeoff between model fit and model complexity. In other words, they answer the question, *is the improvement in fit worth the additional complexity?* AIC, AICc, and BIC all reward models for an improvement in fit, but penalize additional complexity required to do so. Hence, a lower AIC, AICc, and/or BIC score is designed to reflect a “better” model.

```
final_IC_result <- data.frame(rbind(ar1_model$ICs, ar2_model$ICs, ar3_model$ICs,
                                   ma1_model$ICs, ma2_model$ICs, ma3_model$ICs,
                                   arma1_model$ICs, arma2_model$ICs,
                                   arma21_model$ICs, arma12_model$ICs))

rownames(final_IC_result) <- c("AR1 Model", "AR2 Model", "AR3 Model",
                              "MA1 Model", "MA2 Model", "MA3 Model",
                              "ARMA(1, 1) Model", "ARMA(2, 2) Model",
                              "ARMA(2, 1) Model", "ARMA(1, 2) Model")

colnames(final_IC_result) <- c("AIC Score", "AICc Score", "BIC Score")

final_IC_result
```

##	AIC Score	AICc Score	BIC Score
## AR1 Model	-1.269006	-1.268964	-1.245062
## AR2 Model	-1.265786	-1.265703	-1.233861
## AR3 Model	-1.262382	-1.262242	-1.222475
## MA1 Model	-1.268809	-1.268767	-1.244865
## MA2 Model	-1.265734	-1.265650	-1.233809
## MA3 Model	-1.262941	-1.262801	-1.223034
## ARMA(1, 1) Model	-1.265736	-1.265652	-1.233810
## ARMA(2, 2) Model	-1.262113	-1.261902	-1.214224
## ARMA(2, 1) Model	-1.262102	-1.261962	-1.222195
## ARMA(1, 2) Model	-1.262144	-1.262004	-1.222238

By comparing the AIC, AICc and BIC scores between all 5 models, the AR1 model (though marginally so) appears to have the lowest cumulative values, indicating that this is the best fitting model.

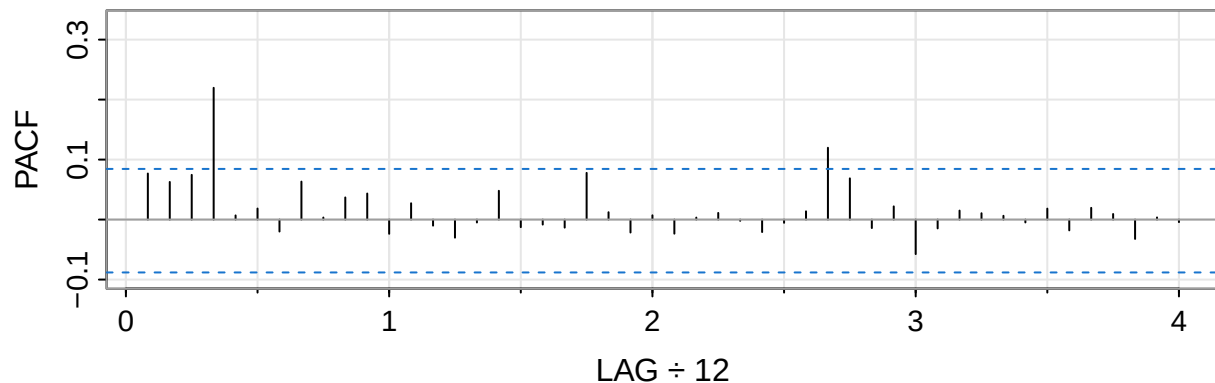
Addressing Potential Heteroskedasticity

In earlier model diagnostic plots, we affirmed that the standardized residuals of these models appear to roughly satisfy the rules of stationarity and constant variance. While the residuals look to be concretely stationary, constant variance is more of a rough visual estimation. Is that variance *truly* constant over time, exhibiting homoskedasticity?

The ACF plot of the squared residuals provides us with an accurate assessment of constant variance. If no significant lags exist, the standardized residuals can be affirmed as having constant variance.

```
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          0.0669 0.0431   1.551  0.1215
## xmean        0.0141 0.0059   2.395  0.0170
##
## sigma^2 estimated as 0.01627598 on 535 degrees of freedom
##
## AIC = -1.269006  AICc = -1.268964  BIC = -1.245062
##
```

Series: Squared Model Residuals



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```
## PACF  0.01  0.01  0.01  0.00  0.02 -0.02  0.02  0.01 -0.03    0  0.00
```

From the ACF and PACF plots above, there seem to be quite a few lags in both that extend beyond the blue significance thresholds, hence exhibiting notable autocorrelation.

While the AR1 model sufficiently accounts for the mean structure of the series, the variance component remains unexplained.

Enhancing Model Fit

To account for the additional heteroskedasticity, a GARCH model could optimize the fit.

GARCH (Generalized Autoregressive Conditional Heteroskedasticity Model): The conditional variance of a time series follows an autoregressive structure, through dependence on previous squared observations and variances.

GARCH Models

```
library(fGarch)

apple_garch <- garchFit(~arma(1, 0) + garch(1, 1), data = apple_data_transformed,
                      cond.dist = "std", trace = FALSE)

summary(apple_garch)
```

```
##
## Title:
##  GARCH Modelling
##
## Call:
##  garchFit(formula = ~arma(1, 0) + garch(1, 1), data = apple_data_transformed,
##    cond.dist = "std", trace = FALSE)
##
## Mean and Variance Equation:
##  data ~ arma(1, 0) + garch(1, 1)
## <environment: 0x116191ee0>
##  [data = apple_data_transformed]
##
## Conditional Distribution:
##  std
##
## Coefficient(s):
##           mu           ar1           omega           alpha1           beta1           shape
## 0.01749491 0.08016021 0.00042942 0.10774474 0.86638710 8.39973009
##
## Std. Errors:
## based on Hessian
##
## Error Analysis:
##      Estimate Std. Error t value Pr(>|t|)
## mu      0.0174949 0.0045158   3.874 0.000107 ***
## ar1     0.0801602 0.0440247   1.821 0.068637 .
## omega   0.0004294 0.0002899   1.481 0.138495
## alpha1  0.1077447 0.0371892   2.897 0.003765 **
## beta1   0.8663871 0.0445335  19.455 < 2e-16 ***
## shape   8.3997301 2.5479606   3.297 0.000978 ***
```

```

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log Likelihood:
## 396.7367      normalized: 0.738802
##
## Description:
## Fri Jun 12 23:14:10 2026 by user:
##
##
##
## Standardised Residuals Tests:
##
##               Statistic      p-Value
## Jarque-Bera Test   R      Chi^2 101.4760440 0.000000e+00
## Shapiro-Wilk Test  R      W      0.9783488 3.835624e-07
## Ljung-Box Test     R      Q(10)  5.3712704 8.650382e-01
## Ljung-Box Test     R      Q(15) 14.6064672 4.801175e-01
## Ljung-Box Test     R      Q(20) 17.7464012 6.041098e-01
## Ljung-Box Test     R^2  Q(10) 10.5692411 3.920469e-01
## Ljung-Box Test     R^2  Q(15) 11.4580378 7.195009e-01
## Ljung-Box Test     R^2  Q(20) 13.7103819 8.448714e-01
## LM Arch Test       R      TR^2  10.0197494 6.142280e-01
##
## Information Criterion Statistics:
##      AIC      BIC      SIC      HQIC
## -1.455258 -1.407369 -1.455504 -1.436524

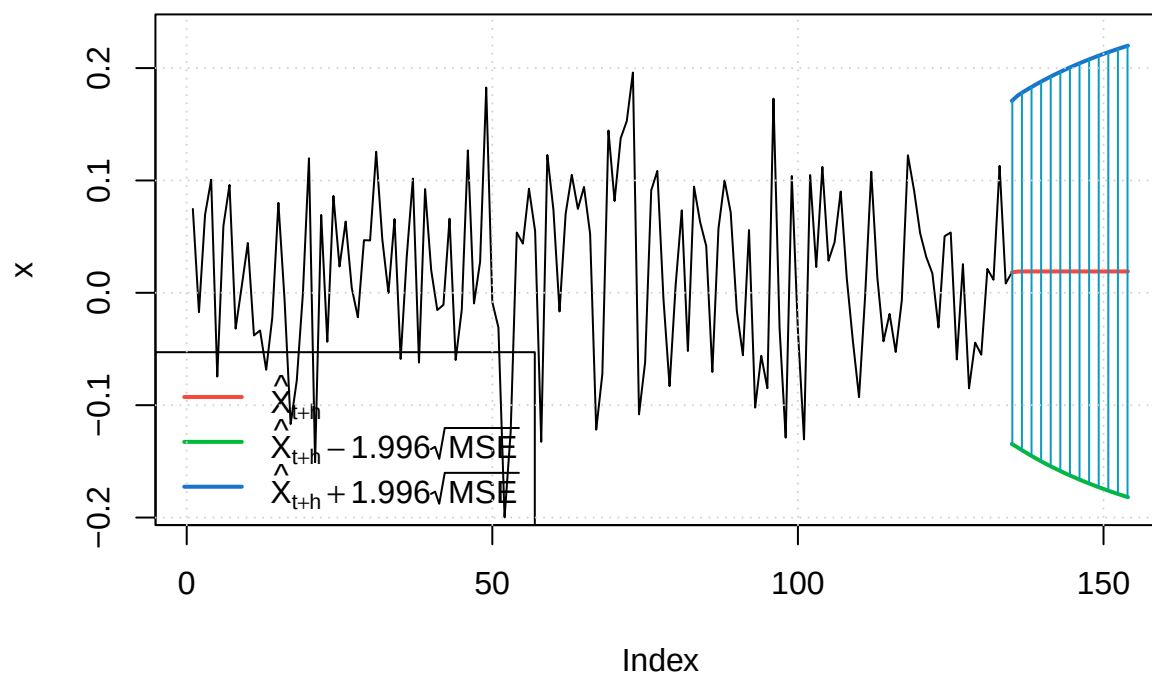
```

We will assess the model fit by looking at the results of the Standardized Residuals Tests. When testing at an alpha level of 0.05, all of the Ljung-Box Test values are far greater than this significance cutoff. Additionally, the LM Arch Test statistic is greater than the chosen alpha level, indicating that there is no evidence of heteroskedasticity remaining in the residuals of the GARCH model.

Hence, we can determine that the optimal model for our data is composed of two parts: an AR1 component for the mean, and a GARCH(1, 1) component for the variance. The final model can be described as an AR1 + GARCH(1, 1) model, or, more formally, an ARIMA(1, 0, 0) + GARCH(1, 1) model.

Forecasting

Prediction with confidence intervals



##	Mean Forecast	Lower Bound	Upper Bound
## May 2025	0.01815860	-0.1345110	0.1708282
## June 2025	0.01895051	-0.1377854	0.1756864
## July 2025	0.01901399	-0.1411545	0.1791825
## August 2025	0.01901907	-0.1444212	0.1824593
## September 2025	0.01901948	-0.1475461	0.1855851
## October 2025	0.01901952	-0.1505352	0.1885742
## November 2025	0.01901952	-0.1533971	0.1914362
## December 2025	0.01901952	-0.1561401	0.1941791
## January 2026	0.01901952	-0.1587714	0.1968105
## February 2026	0.01901952	-0.1612978	0.1993368
## March 2026	0.01901952	-0.1637252	0.2017642
## April 2026	0.01901952	-0.1660592	0.2040982
## May 2026	0.01901952	-0.1683049	0.2063439
## June 2026	0.01901952	-0.1704669	0.2085059
## July 2026	0.01901952	-0.1725495	0.2105885
## August 2026	0.01901952	-0.1745566	0.2125957
## September 2026	0.01901952	-0.1764921	0.2145311
## October 2026	0.01901952	-0.1783592	0.2163983
## November 2026	0.01901952	-0.1801613	0.2182003
## December 2026	0.01901952	-0.1819011	0.2199401

The table above provides Apple's log returns for a forecasting period of 20 months. Due to the fact that the last observable data point was recorded in April 2025, forecasting begins from May 2025.

In essence, this table provides the logged growth or decline rate of Apple stock month-over-month. The *Mean Forecast* column provides the mean estimate, while the *Lower Bound* and *Upper Bound* columns provide the prediction confidence interval lower and upper bounds, respectively.

These predictions can be used in a variety of contexts, including portfolio management, broader national economic reasoning, and corporate planning, among others.

Limitations

While the approaches designed above are robust and defensible, certain limitations still exist. To start with, stocks & financial markets are undoubtedly volatile. The performance of a stock could be dependent on many extraneous factors, some of which are simply not quantifiable.

The modeling methodology mentioned above treats Apple’s log return forecasting as a function of its previous AR and GARCH components. However, it is entirely possible that Apple’s log return forecasting could yield better predictions as being a function of its previous AR and GARCH components, as well as the ARIMA, SARIMA, and/or GARCH components of *other* variable time-series. In other words, Apple’s log return could be better modeled as a function of itself and other variable time-series.

For this, a VAR (Vector Autoregressive Model) would be optimal. Although choosing a good “predictor set” of time-series would take time and effort, this approach could theoretically improve forecasting performance as extraneous factors are now being addressed.

In regards to model fitting, a time-series approach was used throughout this paper. However, more robust approaches, such as supervised machine learning and deep learning, could yield better model fitting performance. While predictor dimensionality would be quite large (time-dependence would need to be translated into lagged predictors), these approaches would theoretically capture even more hidden complexities in the data.

References

Shumway, R. H., & Stoffer, D. S. (2025). Time series analysis and its applications: With R examples (5th ed.). Springer.